

The empirical recurrence for OEIS A234772: recovering a conjecture that is not written down

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7 September 2026

Abstract

OEIS A234772 records “Empirical recurrence of order 52”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 69$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 52, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 52 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A234772 is “Number of $(n+1) \times (3+1)$ 0..4 arrays with 2×2 edge jumps all no more than +1 in one of the clockwise or counterclockwise directions but not both.”. It has offset 1 and begins

2056, 28840, 301776, 4255966, 49077768, 678552840, 8058197452, 109362379520, ...

The entry states:

Empirical recurrence of order 52 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 08:25:41, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 626$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 69$ of the 626, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 69$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 1252$ exact terms of the model returns order 52 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{52} c_i a(n-i),$$

c_1	3	c_{14}	962613692474	c_{27}	−25472310230261259	c_{40}	−835692156481464736
c_2	315	c_{15}	−589329444415	c_{28}	−182128698060288929	c_{41}	230524154960775808
c_3	−666	c_{16}	−11241259302141	c_{29}	69410085164582698	c_{42}	448974968608718592
c_4	−38125	c_{17}	6208104616956	c_{30}	417134684639163840	c_{43}	−112438411886149120
c_5	59855	c_{18}	101544258025851	c_{31}	−151964583996172464	c_{44}	−178613410742936064
c_6	2523094	c_{19}	−51499055357573	c_{32}	−765071404830930528	c_{45}	39187864258001920
c_7	−3064598	c_{20}	−718522528462556	c_{33}	266459723158897032	c_{46}	50527126494373888
c_8	−105715234	c_{21}	339157532755689	c_{34}	1119485601616887620	c_{47}	−9147890031222784
c_9	103056039	c_{22}	4020140428096596	c_{35}	−372015637839894268	c_{48}	−9517749274755072
c_{10}	3035186969	c_{23}	−1783465868110113	c_{36}	−1298669473504021308	c_{49}	1269696565149696
c_{11}	−2457922062	c_{24}	−17906105331129397	c_{37}	409830536695428368	c_{50}	1057336107073536
c_{12}	−62696173869	c_{25}	7518527642856500	c_{38}	1183297333700524952	c_{51}	−78325116567552
c_{13}	43536365994	c_{26}	63780630784284161	c_{39}	−351593270567139376	c_{52}	−51518266933248

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 52, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $52 + 4$ terms removed returns order 52 as well, so $\deg q_{\min} = 52 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $52 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 52 that this sequence satisfies — and that family turns out to have exactly one member.

References

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